Federated Quantum Transformer Architectures for Time-Series Forecasting

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Abstract

Federated quantum transformer architectures use quantum circuits to process sequential data for Machine Learning (ML) pattern recognition in the Noisy Intermediate-Scale Quantum (NISQ) era. They can be used as a tool combining three main ideas: parametrised quantum circuits which act as the structural base for the attention mechanism, the transformer's sequence-modelling layer as the model class, and federated averaging as the training protocol. Unlike static, independently distributed data, time-series data require encoding that preserves their structural properties. This includes temporal dependencies, autocorrelations, and sequential structures within quantum states. As such, model performance directly depends on the federated transformer architecture, specifically the mechanism by which the temporal signal is processed through quantum attention layers and aggregated across decentralised nodes. This work reviews the literature on quantum transformers and federated quantum learning. It focuses on understanding the structural sources of quantum advantage in federated time-series forecasting,by exploring what makes these architectures benefit from a quantum approach. Moreover, it proposes a systematic evaluation study across three datasets with intrinsically distributed structures: the Beijing PRSA air-quality network, the NASDAQ-100 constituent series, and the PEMS-SF traffic occupancy archive.

Keywords: quantum machine learning, federated learning, quantum transformers, time-series forecasting, parametrised quantum circuits, attention mechanisms

1 Introduction

Predicting time-series data is a difficult task across many fields, from climate modelling to finance, transport and grid operations [1]. The classical state of the art is dominated by transformers and their variants, which use attention to weight past observations when predicting the next step [2]. As temporal data becomes more complex, the focus has shifted towards quantum computing in search of solutions with improved performance over classical machine learning methods [3]. Federated quantum transformer architectures, which use decentralised learning and quantum attention mechanisms, show potential for certain learning tasks [4], which demonstrates that variational quantum circuits can be successfully trained across distributed nodes to match centralised performance without ever moving local data.

However, standard quantum models often require data to be gathered in a single location, which creates privacy risks and communication limits [5]. They fail to capture the sequential patterns needed for effective time-series prediction without the use of attention mechanisms [6]. The main question in this study is to determine which structural properties of time-series data allow federated quantum transformers to perform better than classical ones. This requires looking at how data aggregation and the design of quantum attention gates change the way a model learns and behaves [7].

This thesis therefore studies: (i) Which federated aggregation strategies work best for quantum transformer prediction, (ii) How the choice of attention mechanism affects circuit design, such as gate depth and entanglement, and (iii) How to manage the resulting trade-off in accuracy and the costs of the quantum system hardware.

Existing research shows that quantum models can be used for several time-series applications, ranging from climate forecasting [8] and traffic prediction [9] to general temporal pattern recognition tasks. However, it is still unclear when and exactly why federated quantum architectures outperform classical alternatives. Scaling these models is also made difficult by problems like barren plateaus [10] and data heterogeneity constraints [11]. This work reviews different federated aggregation strategies and tries to define the structural factors that create a quantum advantage in federated transformers.

2 Implementation

2.1 Architectural Approaches

Three hybrid quantum-classical architectures are implemented and integrated into a common federated training framework: QuLTSF [12], a from-scratch implementation of the Quantum Self-Attention Neural Network (QSANN) [13], and a time-series adaptation of QuixerTimeSeries [14, 15].

2.1.1 QuLTSF

QuLTSF [16] is an algorithm designed for long-term time series forecasting, by predicting many steps into the future from a long historical window. It replaces the standard linear mapping used in simple forecasting models with a quantum circuit that can improve prediction quality, particularly because quantum circuits can represent complex non-linear transformations with little parameter memory storage.

The model processes each time-series variable independently. For each variable, a classical linear layer compresses the input window of length L into a vector of dimension 2^{n_q}:

$$\mathbf{h} = W_{\text{in}}\,\mathbf{x} + \mathbf{b}_{\text{in}}, \quad W_{\text{in}} \in \mathbb{R}^{2^{n_q} \times L}. \tag{1}$$

That vector is then encoded into n_q qubits via amplitude embedding, which stores the values of $\mathbf{h}$ as quantum state amplitudes:

$$|\psi\rangle = \sum_{i=0}^{2^{n_q}-1} \frac{h_i}{\|\mathbf{h}\|}\,|i\rangle. \tag{2}$$

An ansatz with trainable weights $\boldsymbol{\theta} \in \mathbb{R}^{n_\ell \times n_q \times 3}$ then transforms the state, and all qubits are measured in the Pauli-Z basis to produce a vector $\mathbf{e} \in [-1, +1]^{n_q}$. A final linear layer maps these measurement outcomes to the prediction horizon H. The full model is trained end-to-end via classical backpropagation.

QuLTSF contains no attention mechanism of any kind. It is used in this study as the quantum baseline, establishing what a hybrid quantum-classical model achieves without attention, so that any performance improvement of QSANN or Quixer can be attributed specifically to their attention mechanisms.

2.1.2 QSANN

QSANN was originally proposed for natural language processing, specifically classifying the meaning or sentiment of short sentences. Three adaptations were implemented to move from text classification to time-series forecasting: (i) a classical linear projection maps each multivariate time step $\mathbf{x}_t$ to an angle-encoding vector, (ii) a learnable positional embedding is inserted before the quantum layer to restore temporal order, and (iii) the classification head is replaced with a linear forecasting head.

QSANN is an architecture that produces a attention map, thus showing, for every point in the input sequence, how much the model is focusing on every other point when building its representation. This makes the model easy to analyse in regards to the thesis perspective, which aims to uncover the structural sources of quantum advantage in federated forecasting.

QSANN treats each time step in the sequence as a token. For each token, three separate quantum circuits independently compute a query, a key, and a value. Each circuit encodes the time step's feature values as rotation angles on the qubits, applies a trainable sequence of quantum gates, and then measures the resulting quantum state to produce a short output vector. Once all tokens have their queries, keys, and values, the model computes attention scores classically. The attended representations are then flattened and passed through a linear layer to produce the forecast.

2.1.3 QuixerTimeSeries

The base Quixer model was developed for language modelling. Rather than computing relationships between tokens one pair at a time, it builds a quantum circuit for each time step separately, such that each circuit encodes and transforms that time step's features, and then combines all these circuits into a single quantum state by taking a weighted superposition. The weights are complex numbers that the model learns during training, one per time step, thus learning how much attention to pay to each time step. The result is a single blended quantum state that reflects all the time steps at once, which is then passed through one more quantum processing layer and measured from three different angles. A final linear layer maps these measurements to the forecast.

3 Implementation Layout

All three architectures run as `torch.nn.Module` objects, but QuLTSF and QSANN are built on PennyLane [17], while QuixerTimeSeries uses `torchquantum` [14]. These two libraries have incompatible dependency trees and cannot coexist in the same Python environment, so two separate Conda environments are maintained: one for the PennyLane experiments (QuLTSF and QSANN) and one for the torchquantum experiments (Quixer).

The three datasets are partitioned by their natural distributed structure: the Beijing PRSA archive into one client per monitoring station (twelve clients), the NASDAQ-100 series into one client per constituent asset, and PEMS-SF into one client per sensor. Normalisation is applied per client on that client's own training split only, using z-score scaling. Applying a global scaler would require pooling data across clients at the server, which would violate the federated privacy premise.

4 Current Results

The first experiment evaluates QSANN in a federated setting across the twelve Beijing PRSA monitoring stations over thirty communication rounds. Two configurations are compared: Run 1, with two local training epochs per client per round ($E = 2$), and Run 2, with one local training epoch ($E = 1$). Figure 1 and Table 1 summarise the results.

5 Future Work

The immediate next step is to complete a sixty-round run using the updated training script, which also saves per-station MSE, per-round attention entropy, and a full client drift matrix. Drift was still declining at round 29 in Run 2, so extending training is expected to yield further improvement, with convergence likely around rounds 50–55 based on the current trajectory.

The QuLTSF baseline will be run on all three datasets, providing the no-attention reference point against which QSANN's results are compared. The Quixer experiments, which require the separate `torchquantum` environment, will follow.

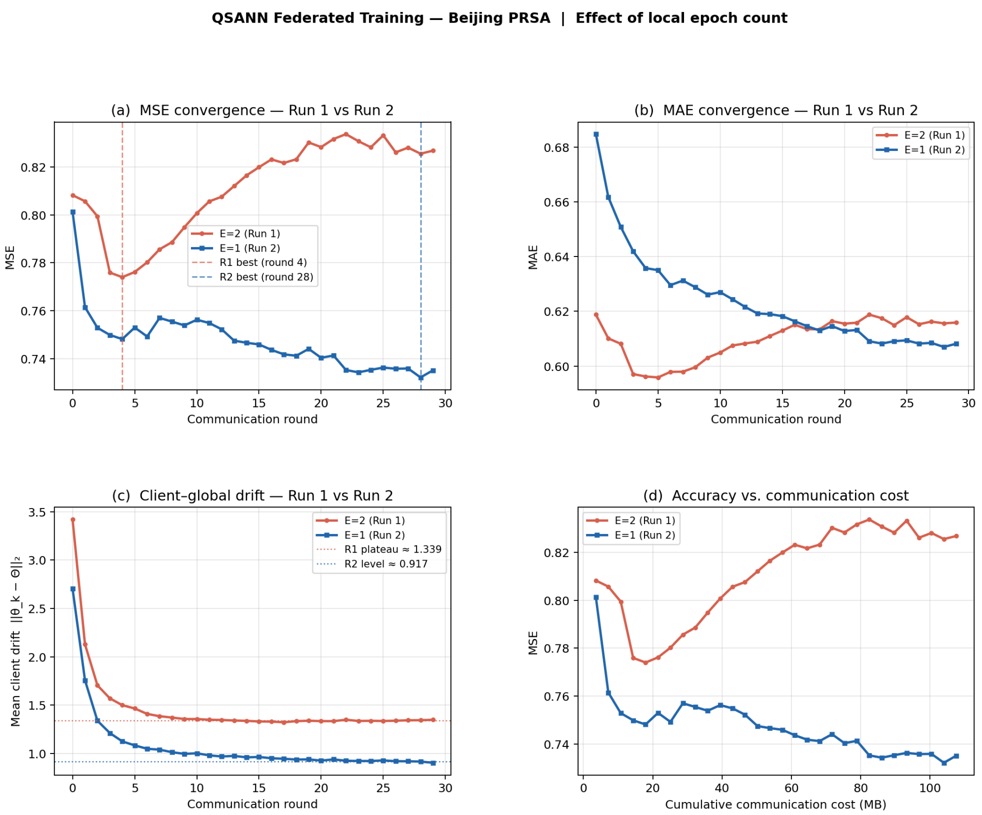

Fig. 1 QSANN Runs Comparison

Table 1 QSANN federated training on Beijing PRSA (12 stations, 30 rounds): comparison of two local-epoch configurations. All error metrics are lower-is-better. Δ = Run 2 − Run 1.

Metric	**Run 1** ($E = 2$)	**Run 2** ($E = 1$)	**Δ (R2 − R1)**
Start MSE	0.80821	0.80126	−0.00695 (↓ better)
Best MSE	0.77400	0.73218	−0.04182 (↓ better)
Best RMSE	0.87977	0.85568	−0.02410 (↓ better)
Final MSE	0.82678	0.73516	−0.09162 (↓ better)
Final RMSE	0.90928	0.85741	−0.05186 (↓ better)
Final MAE	0.61591	0.60818	−0.00773 (↓ better)
Best at round	4	28	—
Drift @ round 29	1.34800	0.90340	−0.44460 (↓ better)
Drift plateau (rounds 10–29)	1.33915	0.94190	−0.39725 (↓ better)
Diverges after best round?	Yes	No	—
Still converging at round 29?	No	Yes	—

References

[1] Lim, B., Zohren, S.: Time-series forecasting with deep learning: a survey. Philosophical transactions of the royal society a: mathematical, physical and engineering sciences **379**(2194) (2021)

[2] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L., Polosukhin, I.: Attention is all you need. In: Advances in Neural Information Processing Systems, vol. 30 (2017)

[3] Trirat, P., Shin, Y., Kang, J., Nam, Y., Na, J., Bae, M., Kim, J., Kim, B., Lee, J.-G.: Universal time-series representation learning: A survey. ACM Computing Surveys (2024)

[4] Chen, S.Y.-C., Yoo, S.: Federated quantum machine learning. Entropy **23**(4), 460 (2021)

[5] McMahan, B., Moore, E., Ramage, D., Hampson, S., Arcas, B.A.: Communication-efficient learning of deep networks from decentralized data. In: Artificial Intelligence and Statistics, pp. 1273–1282 (2017). Pmlr

[6] Cherrat, E.A., Kerenidis, I., Mathur, N., Landman, J., Strahm, M., Li, Y.Y.: Quantum vision transformers. Quantum **8**(arXiv: 2209.08167), 1265 (2024)

[7] Chen, C.-S., Kuo, E.-J.: Quantum adaptive self-attention for quantum transformer models. arXiv preprint arXiv:2504.05336 (2025)

[8] Hsu, Y.-C., Li, T.-Y., Chen, K.-C.: Quantum kernel-based long short-term memory. In: 2025 IEEE International Conference on Acoustics, Speech, and Signal Processing Workshops (ICASSPW), pp. 1–5 (2025). IEEE

[9] Schetakis, N., Bonfini, P., Alisoltani, N., Blazakis, K., Tsintzos, S.I., Askitopoulos, A., Aghamalyan, D., Fafoutellis, P., Vlahogianni, E.I.: Data re-uploading in quantum machine learning for time series: application to traffic forecasting. arXiv preprint arXiv:2501.12776 (2025)

[10] Larocca, M., Czarnik, P., Sharma, K., Muraleedharan, G., Coles, P.J., Cerezo, M.: Diagnosing barren plateaus with tools from quantum optimal control. Quantum **6**, 824 (2022)

[11] Zhao, Y., Li, M., Lai, L., Suda, N., Civin, D., Chandra, V.: Federated learning with non-iid data. arXiv preprint arXiv:1806.00582 (2018)

[12] Chittoor, H.H.S., Griffin, P.R., Neufeld, A., Thompson, J., Gu, M.: Qultsf: Long-term time series forecasting with quantum machine learning. arXiv preprint arXiv:2412.13769 (2024)

[13] Li, G., Zhao, X., Wang, X.: Quantum self-attention neural networks for text

classification. Science China Information Sciences **67**(4), 142501 (2024)

[14] Park, J.J., Seo, J., Bae, S., Chen, S.Y.-C., Tseng, H.-H., Cha, J., Yoo, S.: Resting-state fmri analysis using quantum time-series transformer. In: 2025 IEEE International Conference on Quantum Computing and Engineering (QCE), vol. 1, pp. 2352–2363 (2025). IEEE

[15] Khatri, N., Matos, G., Coopmans, L., Clark, S.: Quixer: A Quantum Transformer Model. arXiv:2406.04305 (2024)

[16] Zeng, A., Chen, M., Zhang, L., Xu, Q.: Are transformers effective for time series forecasting? In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 37, pp. 11121–11128 (2023)

[17] Bergholm, V., Izaac, J., Schuld, M., Gogolin, C., Ahmed, S., Ajith, V., Alam, M.S., Alonso-Linaje, G., AkashNarayanan, B., Asadi, A., et al.: Pennylane: Automatic differentiation of hybrid quantum-classical computations. arXiv preprint arXiv:1811.04968 (2018)